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Instructions to Convert C8.3G to a C Gas Plus

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Preparatory Steps

After the completion of this upfit, the engine dataplate **must** be stamped with the new Control Parts List (CPL). This engine modification **must** be reported to Cummins± on Form 1877 "Notice of Engine Modification", which can be found in the Warranty Administration Manual.

Refer to the C8.3G, C Gas Plus, and L Gas Plus Engines Base Engine Trouble Shooting and Repair Manual, Bulletin 3666206, and Trouble Shooting and Repair Manual Electronic Control System Gas Plus Engines, Bulletin 4021317 for the specific procedures related to the following conversion steps.

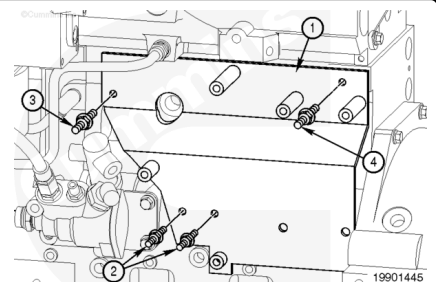


Pistons

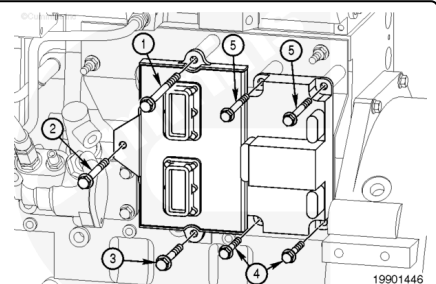
Remove the 6 pistons and rings from the engine. Install the new design pistons, Part Number 3992118, with the new rings, Part Number 3607359 - top, 3992149 - 2nd, 3992150 - oil ring, all of which come in the piston kit, Part Number 4089346. The new piston rings utilize a 3-piece oil ring design for improved oil consumption.

Electronic Control Module (ECM)

Remove the old ECM, Part Number 3965953, ignition module, wire harness and bracket, Part Number 3934694. Replace the oil fill capscrews with M8-M8 studded capscrews (2), Part Number 3941976. Attach the bracket (1), Part Number 3934694, using a M10-M8 studded capscrew (3), Part Number 3944152, in the upper left corner, and an M10 x 1.5 x 20 capscrew (4), Part Number 3903990, in the upper right corner. Secure the bracket with two of the included nuts.

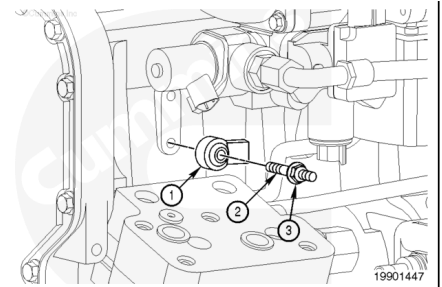


Attach the ECM, Part Number 3965953, using the M8 x 1.25 x 110, M8 x 1.25 x 70 and M8 x 1.25 x 50 capscrews (1,2,3), Part Numbers 3927213, 3903096 and 3900633. The old ignition module will be reused and attached using the M8 x 1.25 x 90 (5) and M8 x 1.25 x 25 capscrews (4), Part Number 3913366 and Part Number 3900631.

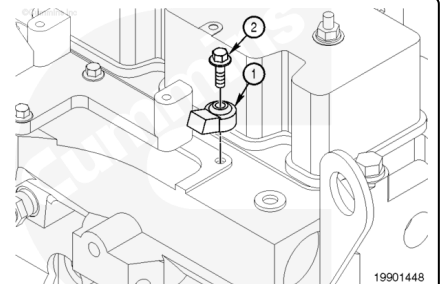


Knock Sensors

Two knock sensors, Part Number 3607945, are used. One is installed on the cylinder block behind the fuel control housing (1) with knock sensor stud (2), Part Number 3938841, and an M8 nut (3), Part Number 3883245.



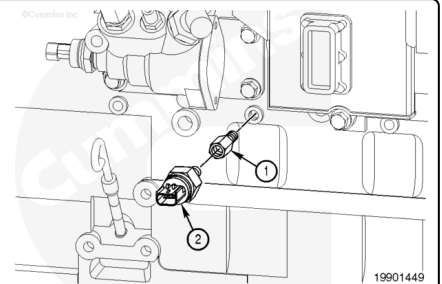
The other knock sensor (1) is installed above cylinder number 6 with an M8 capscrew (2), Part Number 3900632. Angle the sensor connector outward to allow for the engine harness connection. Confirm the ignition coil part numbers are 3608003, otherwise replace the ignition coils. Connect the new ignition coil harness, Part Number 3934686, which is longer so that it can reach the new location of the ignition module.



Oil Pressure and Temperature Sensor

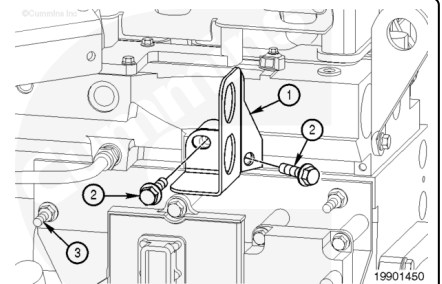
Attach the oil rifle connector (1), Part Number 3607279, to the forward pipe thread hole.

Connect the oil temperature and pressure sensor (2), to the connector.

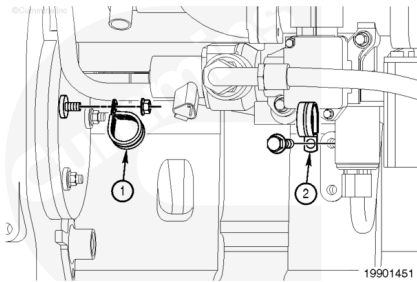


Engine Harness

Attach the engine harness retainer bracket (1), Part Number 3937309, in the location shown with two M12 x 1.5 x 20 capscrews (2), Part Number 3922741. The OEM connectors will attach to the bracket with nuts, Part Number 3617958, and washers, Part Number 3617959. Connect the engine harness, Part Number 3968266, to the ECM and ignition module. P-Clips are connected to the harness for mounting to the engine. The connectors on the harness are keyed so they can **not** be plugged into the incorrect sensor. The grounding strap of the harness should be attached to the studded capscrew (3) on the upper left corner of the ECM bracket with an M8 nut, Part Number 3883245. The upper sensor in the intake manifold will **not** be used.

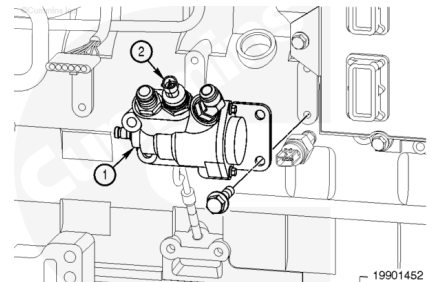


The engine harness is attached with P-clips. One attaches to a capscrew on the gear housing (1) and another connects to a capscrew below the fuel control housing (2).



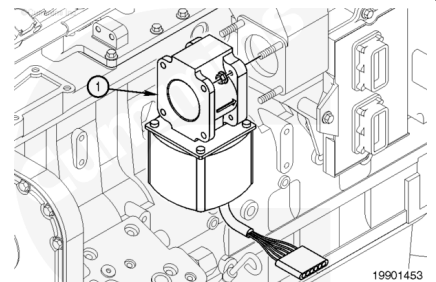
Fuel Pressure Sensor, Primary

The hole in the old fuel pressure regulator housing, Part Number 3606651, is the wrong thread size for the new sensor. Remove the fuel pressure regulator and move the pressure regulator valve, Part Number 3607278, and fittings to the new pressure regulator housing (1), Part Number 3938821. Insert fuel pressure sensor, Part Number 3330527, in the hole (2). Install the new pressure regulator housing, Part Number 3938821, to the engine.



Throttle Actuator

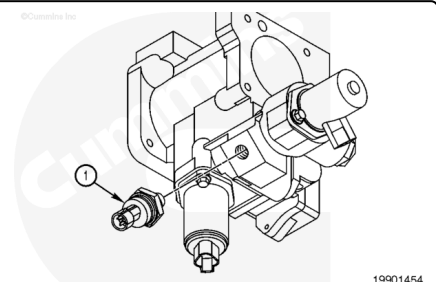
Remove the air intake elbow, fuel control housing, Part Number 3929939, and throttle actuator, Part Number 3928273. Install the new throttle actuator (1), Part Number 3970806, on the intake manifold.



Fuel Pressure Sensor and Temperature Sensor, Secondary

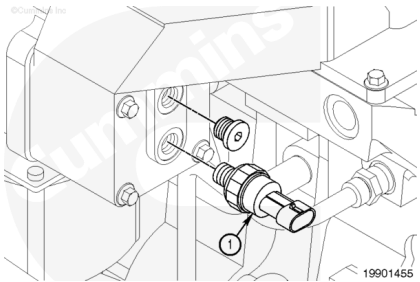
Note : If the housing already has a sensor hole machined but plugged, don't replace the housing. Install the sensor in place of the plug.

Transfer the Venturi, Part Number 3928282, from inside the old housing to the new one along with the gas mass flow sensor, Part Number 3929935, fuel shut-off valve, Part Number 3929160, fuel control valve, Part Number 3928310, and waste gate control valve, Part Number 3928342. Install the fuel pressure and temperature sensor (1), Part Number 3417195, into the new fuel control housing, Part Number 3934703. Install the new fuel control housing, Part Number 3934703, in place of the old housing and reattach other components.



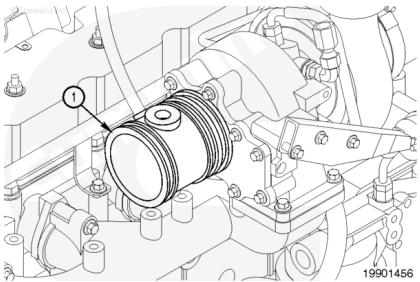
Intake Manifold Pressure and Temperature Sensor

The C Gas Plus engine uses an intake manifold pressure and temperature combination sensor (1), Part Number 4009913, instead of two separate sensors. Remove the lower sensor, and install the combination sensor. Leave the upper sensor in place as a plug (or replace with a plug). The engine harness will **not** connect to it.



Exhaust Back Pressure

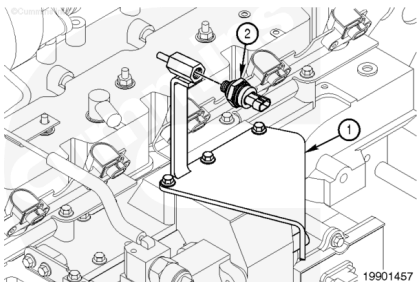
Remove the turbocharger outlet connection. Install one of the turbocharger outlet connections (1) for the C Gas Plus engine based upon the engine application. Consult the following chart for outlet part number details.



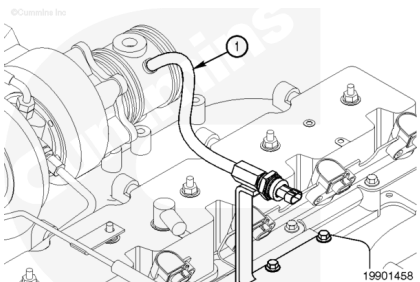
Turbocharger Outlet Connection

C8.3G Part Number	C Gas Plus Part Number
3928293	3938669
3930101	3938668
3933050	3938667

Remove the closed crankcase filter bracket, Part Number 3933515. Install the closed crankcase filter bracket, Part Number 3607282 (1). Install the back pressure sensor (2), Part Number 3348579, into the hole on the closed crankcase filter bracket, Part Number 3607282.



The selection of the hose-tube assembly (1) used to connect the back pressure sensor, Part Number 3348579, and the hole in the turbocharger outlet connection, are based upon engine application. Maintain a downward slope away from the sensor to allow condensation to drain out. Consult the table below for pressure sensing tube assembly information.



Pressure Sensing Tube Assembly on C Gas Plus				
	Turbocharger Outlet Connection			
Turbocharger	3938668, 3938667	3938669, Pointing Up	3938669, Pointing Down	3938669, Horizontal
HMRO	3607325	3938319	N/A	3607325
LMRO	3607352	3938323	3938331	3607352
HMFO	3607349	3607348	3938325	3607349
LMFO	3607351	3938321	3938329	3607351

Conversion Parts List

Part Number	Part Description	Quantity
3965953	ECM	1
3934694	Bracket, ECM	1
3937309	Bracket, wiring retainer	1
3607279	Connector, oil rifle	1
3607282	Bracket, filter	1
3970806	Actuator	1
3968266	Harness, wiring	1
3934686	Harness, ignition	1
3992118	Piston	6
3607359	Compression ring, piston	6
3992149	Compression ring, piston	6
3992150	Ring, piston oil control	6
3607349	Tube, pressure sensing - High Mount Front Out (HMFO) - SF Assembly	*
3607325	Tube, pressure sensing - High Mount Rear Out (HMRO) Assembly	*

3607351	Tube, pressure sensing - Low Mount Front Out (LMFO) Assembly	*
3607352	Tube, pressure sensing - Low Mount Rear Out (LMRO) Assembly	*
3607348	Tube, pressure sensing - High Mount Front Out (HMFO) SF Assembly - Up	*
3938319	Tube, pressure sensing - High Mount Rear Out (HMRO) Assembly - Up	*
3938321	Tube, pressure sensing - Low Mount Front Out (LMFO) Assembly- Up	*
3938323	Tube, pressure sensing - Low Mount Rear Out (LMRO) Assembly - Up	*
3938325	Tube, pressure sensing - High Mount Front Out (HMFO) Assembly - Down	*
3938329	Tube, pressure sensing - Low Mount Front Out (LMFO) Assembly - Down	*
3938331	Tube, pressure sensing - Low Mount Rear Out (LMRO) Assembly - Down	*
3938841	Stud, knock sensor	1
3900632	Capscrew (M8 x 1.25 x 35)	1
3938669	Connection, exhaust outlet	*
3938668	Connection, exhaust outlet	*
3938667	Connection, exhaust outlet	*
3934703	Housing, fuel control	1
3938821	Housing, pressure regulator	1
4009913	Sensor, intake manifold combination	1
* Select 1 pressure sensing tube and 1 exhaust outlet connection compatible to the turbocharger orientation.		

Holds Bracket to Block		
3941976	Capscrew, studded M8 - M8	2
3944152	Capscrew, studded M10 - M8	1
3883245	Nut M8	4
Holds ECM to Bracket		
3903990	Capscrew, M10 x 1.5 x 20	1
3900633	Capscrew, M8 x 1.25 x 50	1
3903096	Capscrew, M8 x 1.25 x 70	1
3927213	Capscrew, M8 x 1.25 x 110	1
Holds Ignition Module to Bracket		
3900631	Capscrew, studded M8 x 1.25 x 25	2
3913366	Capscrew, studded M8 x 1.25 x 90	2
Holds OEM Connector Bracket to Block		
3922741	Capscrew, studded M12 x 1.5 x 20	2
Other Parts		
3330527	Sensor, fuel pressure	1
3417195	Sensor, fuel, oil, pressure and temperature	2
3607945	Sensor, knock	2
3348579	Sensor, back pressure	1
3617959	Washer, OEM connector	2

Wiring Conversion

Original Equipment Manufacturer (OEM) Wiring Conversion (C8.3 G to C Gas Plus)			
WW J2	21 Pin	C+	Function
4		23	Remote power take-off (PTO)
12		13	Door interlock
16		G	Pedal supply 5 VDC
3		X	Pedal position
20		15	Idle OFF
13		14	Idle ON

Original Equipment Manufacturer (OEM) Wiring Conversion (C8.3 G to C Gas Plus)			
WW J2	21 Pin	C+	Function
8	A	28	1587 +
15	F	29	1587 -
17		J	Pedal return
21		9	Cruise ON or OFF
6		7	Cruise resume
5		8	Cruise set
11		17	Brake switch
11		16	Clutch switch
		24	Malfunction override
22		5	VSS +
23		6	VSS -
		1	Red lamp
		2	Yellow lamp
		3	White lamp
		21	Diagnostic switch
		B	12 VDC 5 Ampere
		C	12 VDC 5 Ampere
		D	12 VDC 5 Ampere
		M	12 VDC 5 Ampere
		E	Ground
		F	Ground
		H	Ground
		P	Ground
	G	A	Key 5 Ampere
	G	25	Key 10 Ampere
	G	26	Key 10 Ampere
	D	27	Key 15 Ampere
	D	30	Fused with Pin 27

Note : Refer to Wiring Diagram, Bulletin 4021276.

Document History

Date	Details
14-03-2017	Module Converted

Date	Details
2017-4-4	Update to Format

Last Modified: 04-Apr-2017
